

Wideband Waveform Design Principles for Solid-State Weather Radars

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ABSTRACT

The use of solid-state transmitters is becoming increasingly viable for atmospheric radars and is a key part of the strategy to realize any dense network of low-powered radars. However, solid-state transmitters have low peak powers and this necessitates the use of pulse compression waveforms. In this paper frequency diversity in a wideband waveform design is proposed to mitigate the low sensitivity of solid-state transmitters. In addition, the waveforms mitigate the range-eclipsing problem associated with long pulse compression. An analysis of the performance of pulse compression using mismatched compression filters designed to minimize sidelobe levels is presented. The impact of the range sidelobe level on the retrieval of Doppler moments is discussed. Realistic simulations are performed based on both the Colorado State University–University of Chicago–Illinois State Water Survey (CSU–CHILL) and the Center for Collaborative Adaptive Sensing of the Atmosphere (CASA) Integrated Project I (IP1) radar data.

1. Introduction

Weather radar systems using solid-state transmitters are becoming increasingly viable. The transition from traditional high-powered transmitters to solid-state transmitters is also part of the strategy to realize a network of low-cost electronically steered X-band radars. **However, solid-state transmitters have low peak powers, which degrade the sensitivity of the radar if they are used in a conventional way with a narrow transmit pulse.** Sensitivity requirements with low peak power transmitters necessitate the use of pulse compression waveforms. Pulse compression radars transmit long pulses to achieve adequate sensitivity and range resolution. Pulse compression has been in use with hard target radars for several decades and significant advances have been made in its technology and implementation (Cook and Bernfeld 1993). Pulse compression techniques for hard target radar systems are well documented in literature (Skolnik 1990). Their applicability to weather radar, however, is relatively rare. Typically weather radar targets are extended volume scatterers

and range sidelobes are a major source of error for quantitative applications (Mudukutore et al. 1998). Mismatched filtering techniques to improve the range sidelobes have been studied, keeping in mind their applicability to weather radar.

Although pulse compression waveforms provide adequate sensitivity they have a major drawback in providing coverage at close range. Pulse compression waveforms suffer from a blind zone that occurs because the receiver does not receive any signal while the long pulse is being transmitted. The transmission of long pulses is necessary to achieve adequate sensitivity but results in blind ranges. One solution for using pulse compression waveforms to overcome the blind range problem is by staggering short and long pulses. The long pulse provides adequate sensitivity at farther ranges and the short pulse provides coverage in the blind range region. However, alternating between long and short pulses reduces maximum unambiguous velocity. A pulse train of short pulses followed by a pulse train of long pulses can be used to provide adequate sensitivity and coverage in the blind range region. However, such a pulsing scheme will increase the dwell time, resulting in a slower scan speed.

In this paper a class of frequency-diverse wideband waveforms is presented to mitigate both the low sensitivity

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of solid-state transmitters and the blind zone problem associated with pulse compression. The proposed waveform is designed and implemented for a dual-polarization X-band radar operating in simultaneous transmit and receive (STAR) mode. Frequency diversity is viable because solid-state transmitters can achieve a much wider bandwidth with acceptable efficiency. The proliferation of low-cost advanced digital processors, as well as the advances in the digital transmitter control technology and low-power solid-state transmitter technology, makes the class of frequency-diverse wideband waveforms viable for implementation. Some of the important considerations that need to be taken into account for weather radars using pulse compression are described in George et al. (2008).

This paper is organized as follows: Section 2 provides the relation between transmit waveform and sensitivity to develop a “sensitivity-mapped generalized waveform.” The nonlinear frequency modulation (NLFM) pulse compression waveform, its parameterization, and its associated compression filter design is presented in section 3. The sidelobe characterization of the NLFM pulse compression waveform and mismatched filter are described in section 4. The performance of the NLFM pulse compression waveform in the retrieval of the spectral moments and polarimetric variables for volume targets is described in section 5. A frequency-diverse pulse compression waveform that mitigates the low sensitivity of the solid-state transmitter radars and mitigates blind range issues of pulse compression waveform is presented in section 6. The important conclusions are summarized in section 7.

2. Sensitivity-mapped generalized waveforms

The equivalent reflectivity factor of a weather radar is estimated from the received power at the output of the antenna. The equivalent reflectivity factor is given by (Bringi and Chandrasekar 2001)

$$Z_e = C' \bar{P}_{\text{ref}} R^2. \quad (1)$$

In the above equation $\bar{P}_{\text{ref}}$ is the received power at the reference antenna port, R is the radar range to the resolution volume of the observation, and C' is the radar constant given as

$$C' = \frac{1}{\pi^5 |K_w|^2} \left(\frac{2}{c\tau} \right) \left[\frac{(4\pi^3 l_{\text{wg}}^2)}{P_t G_0^2} \right] \left(\frac{8 \ln 2}{\pi \theta_B \phi_B} \right) \lambda^2. \quad (2)$$

In the above equation K_w is the dielectric factor of water, τ is the pulse width, P_t is the peak transmit power, G_0/l_{wg} is the antenna gain, including the waveguide loss (l_{wg}), θ_B and ϕ_B are the 3-dB beamwidths in elevation and azimuth, respectively, and λ is the operating wavelength of

the radar. In practice Z_e is expressed in $\text{mm}^6 \text{m}^{-3}$ (dBZ in decibel scale). The radar equation can now be written as

$$Z_e(\text{dBZ}) = \bar{P}_{\text{ref}}(\text{dBm}) + C(\text{dB}) + 20 \log[R(\text{km})], \quad (3)$$

where the radar constant C is given by

$$C = 10 \log \left\{ \frac{1}{\pi^5 |K_w|^2} \left(\frac{2}{c\tau} \right) \left[\frac{(4\pi^3 l_{\text{wg}}^2)}{P_t G_0^2} \right] \left(\frac{8 \ln 2}{\pi \theta_B \phi_B} \right) \lambda^2 10^{21} \right\}, \quad (4)$$

where the units conversion factor 10^{21} is the constant necessary to express the reflectivity ($\text{mm}^6 \text{m}^{-3}$). It is customary to define the minimum detectable reflectivity at a specified range when the signal-to-noise ratio (SNR) is 0 dB (Bringi and Chandrasekar 2001). Therefore, the sensitivity of the radar is studied in terms of the minimum detectable reflectivity. The sensitivity is governed by the radar constant given in (4) and the receiver equivalent noise bandwidth, which dictates the amount of noise power in the received signal. For a given radar the minimum observable reflectivity factor is directly proportional to the receiver bandwidth and inversely proportional to the product of pulse width and peak power. Conventional weather radars operating at S band, such as Weather Surveillance Radar-1988 Doppler (WSR-88D) and Colorado State University–University of Chicago–Illinois State Water Survey (CSU–CHILL), transmit a 1- μs pulse with peak power in excess of 500 kW. However, solid-state transmitters have low peak power and are generally on the order of tens or hundreds of watts. Therefore, in order to attain adequate sensitivity, a much longer pulse must be transmitted. One of the drawbacks of improving sensitivity solely by transmitting long pulses is the subsequent degradation of range resolution. The resolution of the measurements along range is a function of the transmit pulse width. Pulse compression techniques using modulated long pulses have been used to improve sensitivity and achieve range resolution that is similar to that of a short pulse (Skolnik 1990).

Pulse compression waveforms enable the use of low peak power transmitters to achieve adequate sensitivity, but these waveforms have extended blind ranges or blind zones. The inherent problem of using a single antenna is that the antenna is not used in receive mode while the waveform is being transmitted. Therefore, there is no received signal for the duration of the time while the transmission is active. This results in a blind range (either a blind zone or range eclipsing) for the radar where there are no observations available. The problem of a blind range has been addressed by alternating short and long pulses (Skolnik 1990). Such solutions have been

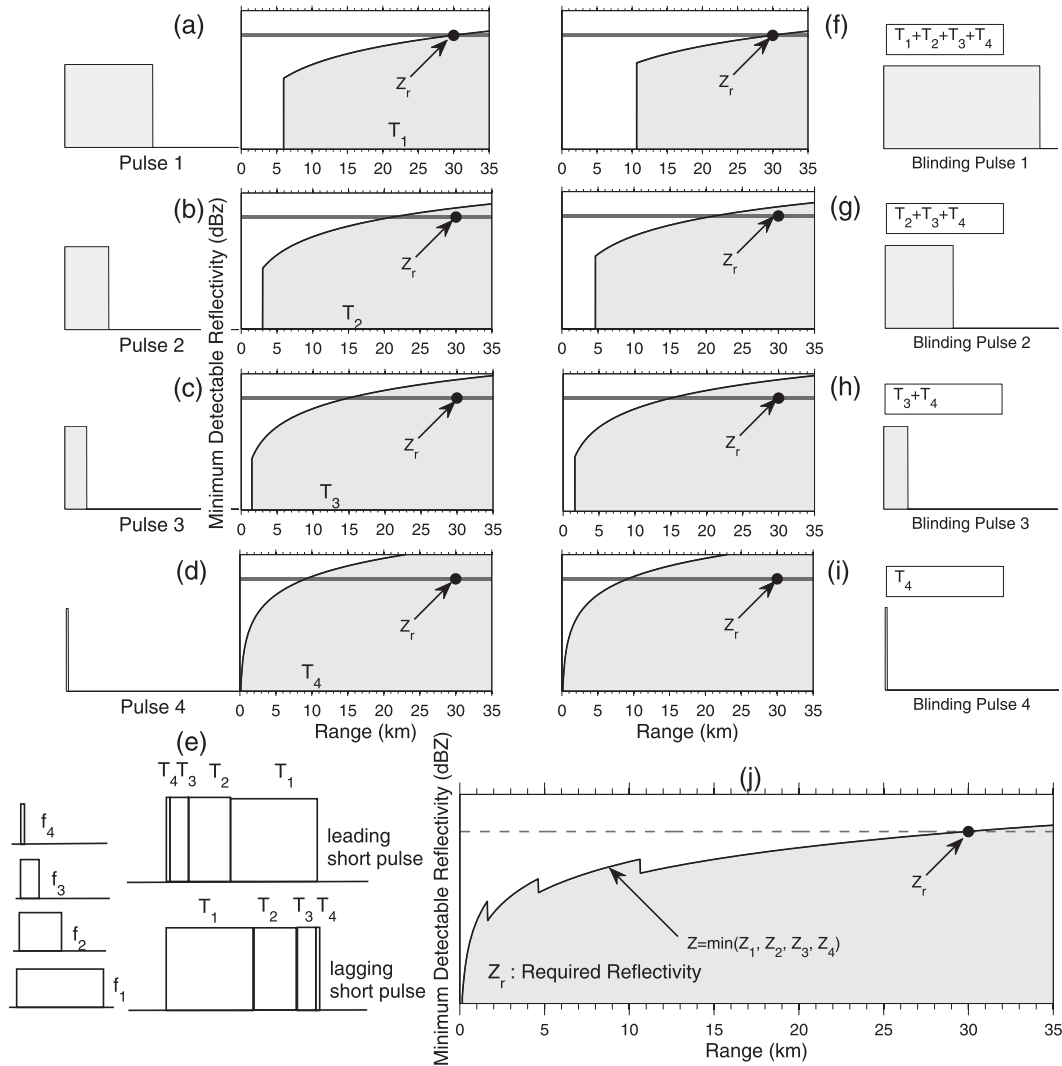


FIG. 1. Minimum detectable reflectivity of a sensitivity-mapped frequency diversity waveform with many subpulses.

implemented for weather radars using pulse compression (Nakagawa et al. 2005). However, alternating between short and long pulses typically leads to two drawbacks. First, if the long and short pulses are switched on a pulse-to-pulse basis, then the maximum unambiguous velocity is lowered because the pulse repetition time (PRT) is doubled. Second, if a pulse train of short pulses is alternated with a pulse train of long pulses, then the dwell times are made much larger.

In this section a sensitivity-mapped generalized waveform is proposed that uses frequency diversity to mitigate for a blind range and a pulse compression waveform to negate the impact of the low peak power of solid-state transmitters. Design principles are presented where sensitivity is mapped to the transmitted pulse width, which enables the design of generalized transmit waveforms. The transmission of a long pulse provides high sensitivity.

Figure 1a shows an illustration of achieving the required sensitivity (Z_r) with a long pulse T_1 . The long pulse T_1 achieves the high sensitivity at the expense of a large blind range, as shown in Fig. 1a. The blind range can be reduced by lowering the pulse width, as shown in Figs. 1b,c. Lowering the pulse width degrades the sensitivity and does not achieve the minimum required sensitivity Z_r , as shown in Figs. 1b,c. However, observations closer to the radar within the blind range of T_1 are made with adequate sensitivity. As the pulse width becomes smaller the blind range becomes smaller, but it results in degraded sensitivity. The observations close to the radar can be made with a short pulse. The sensitivity of the short pulse is inadequate at farther ranges, as shown in Fig. 1d. It is also important to observe that the sensitivity of the short pulse is limited in the upper end of the blind zone of the longest pulse T_1 . Both sensitivity and blind range mitigation can

be obtained if observations from all of the pulses are available at the same time. Observations from all of the pulses can be obtained at the same time by using frequency diversity. The pulses can be transmitted such that their center frequencies are separated with nonoverlapping frequency bands. Such a transmission can be viewed as making measurements with multiple radars in parallel at identical time instants with independent pulses. For example, Fig. 1e shows four independent pulses at frequencies f_1, f_2, f_3 , and f_4 that could be transmitted simultaneously. However, implementation of such a waveform with multiple transmitters is not practical, and a transmission of such a waveform with a single transmitter is not useful because of the intermodulation effects.

The generalized sensitivity-mapped waveform can be realized practically when the set of pulses at different center frequencies are multiplexed in time, as shown in Fig. 1e. Frequency diversity is viable because solid-state transmitters can achieve a much wider bandwidth with acceptable efficiency. In addition, the electronically scanned (phase-phase steering) will employ solid-state transmit-and-receive (T/R) modules, making it critical to design waveforms for such systems. The order of the subpulses in the generalized waveform is important because the blind range for a given subpulse (T_k) depends on the subpulses leading it. All of the subpulses leading a given subpulse together form a blinding pulse. Figure 1e shows the generalized waveform with two of the many ways to multiplex the transmitted subpulses in time—one with a lagging short pulse and another with a leading short pulse. It can be observed that the largest blinding pulse occurs for a leading short pulse and similar blind pulses applied to the shorter pulses leading longer pulses, as seen in Fig. 1e. Waveforms with leading short pulses do not mitigate the sensitivity and blind range problem. However, multiplexed subpulses with lagging shorter subpulses with a lagging short pulse, as shown in Fig. 1e, will enable the mitigation of a blind range and achieve the required sensitivity.

For a generalized transmit waveform, the complex envelope can be written as a sum of N components. To be able to avoid contamination of echoes from these individual components from the various ranges, the receiver must be able to separate the received signals corresponding to each component. This is accomplished by frequency diversity, where each subpulse is at a different center frequency. The longer subpulses use pulse compression to obtain adequate range resolution. The transmit pulse is a frequency-diverse pulse compression waveform. The complex envelope of the generalized waveform is given by

$$g(t) = \sum_{i=1}^N g_i(t - T_{i-1}) e^{j2\pi f_i(t - T_{i-1})}, \quad (5)$$

$$T_1 > T_2 > T_3 > \dots > T_N, \quad \text{and} \quad (6)$$

$$\sum_{i=1}^N T_i \leq T_{\max}, \quad (7)$$

where $g_i(t)$ is the complex envelope of the i th component of the waveform at a frequency offset f_i , T_i is the pulse width of each component with $T_0 = 0$, and $T_{\max}$ is the maximum transmit pulse length limited by the hardware. The blinding pulse for each subpulse $T_{b,i}$ is given by

$$T_{b,i} = \sum_{j=i}^N T_j. \quad (8)$$

The blind range at each frequency is $cT_{b,i}/2$. Figure 1f shows an illustration of the blind range for the longest pulse (T_1), which corresponds to the blinding pulse $T_1 + T_2 + T_3 + T_4$. Although T_1 has a large blind range it provides the required sensitivity (Z_1) at farther ranges. Subpulses T_2 and T_3 provide slightly degraded sensitivities (Z_2 and Z_3) in the blind range of T_1 , as shown in Figs. 1g,h. The blind ranges for T_2 and T_3 are smaller and correspond to $T_2 + T_3 + T_4$ and $T_3 + T_4$, respectively. The lagging short-pulse T_4 provides adequate sensitivity (Z_4) close to the radar without any blind range, as shown in Fig. 1i.

The minimum observable reflectivity Z_i obtained with subpulses with varying sensitivities and blind ranges is shown in Figs. 1f–i. The observed sensitivity obtained with the frequency-diverse pulse compression waveform is given by

$$Z_{\min}(r) = \min[Z_1(r), Z_2(r), \dots, Z_N(r)]. \quad (9)$$

The minimum observable reflectivity for $N = 4$ is shown in Fig. 1j, and it is clearly seen that the required sensitivity is achieved without any blind range. In a high-powered radar system $Z_{\min}(r) \propto r^{-2}$, but the sensitivity of radars using solid-state transmitters with complex transmit pulse cannot be characterized by $Z_{\min}(r) \propto r^{-2}$. The sensitivity profile is governed by the frequency-diverse wideband transmit waveform. The sensitivity is mapped to the individual subpulses ($T_1, T_2, \dots, T_N$), as shown in Fig. 1j with $N = 4$. The shortest pulse (T_N) and the longest pulse (T_1) are mapped based on the sensitivity requirements at far (Z_r) and close ranges, respectively (see Fig. 1). As can be observed in Fig. 1j, the sensitivity is a piecewise function of range and provides adequate sensitivity for the entire coverage region of the radar.

The receiver used to support the frequency-diverse waveform is complex because multiple receive channels have to be implemented on the digital receiver. A simpler receiver with two intermediate frequency (IF) channels might be considered to simplify the digital receiver

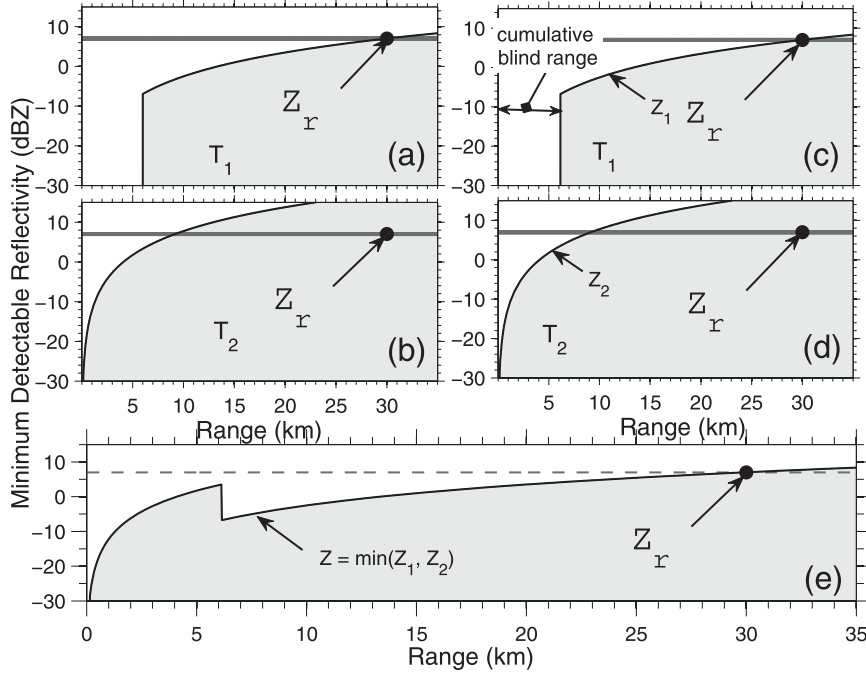


FIG. 2. Minimum detectable reflectivity of a sensitivity-mapped frequency diversity waveform with two subpulses.

design. For a simple receiver with two IF channels, the frequency-diverse pulse compression waveform reduces to a long-pulse T_1 (Fig. 2a) and short-pulse T_2 (Fig. 2b) stacked together. The two subpulses (Figs. 2c,d) provide observations without blind range and required sensitivity at far ranges, as shown in Fig. 2e. However, the sensitivity at closer ranges is degraded because observations are made with the short pulse beyond ranges where the short pulse provides adequate sensitivity. Therefore, more than two subpulses ($N > 2$) are more likely required to obtain adequate sensitivity over the entire coverage region of the radar. The number of subpulses is limited by the duty cycle of the transmitter and the operating bandwidth of the receiver.

3. Pulse compression waveforms

A basic pulse compression waveform consists of a coding signal that modulates the transmitted pulse. Phase and frequency modulation (FM) have been widely used for pulse compression applications. For a pulse compression waveform the complex envelope of the transmitted wideband pulse is given by

$$g(t) = u(t) \exp \left[j2\pi \int_{-T/2}^t f(\tau) d\tau \right], \quad (10)$$

where $u(t)$ and T are the envelope and length of the transmitted pulse, respectively, and $g(t)$ is the complex envelope

of the transmitted pulse. The frequency modulation is governed by $f(t)$. The matched filter, which maximizes the signal-to-noise ratio, is obtained as $g^*(-t)$, where $*$ indicates complex conjugate. The matched filter is completely determined by the complex envelope of the transmit pulse.

a. Nonlinear frequency modulation

The FM characteristic similar to the waveform described by Griffiths and Vinagre (1994) is proposed for pulse compression. The FM characteristic is decomposed into linear and nonlinear component as

$$f(t) = \begin{cases} \frac{B}{T} \left(\frac{1 - k_B}{1 - k_T} \right) t & |t| \leq T(1 - k_T) \\ \varphi(t) \text{sgn}(t) & |t| > T(1 - k_T) \end{cases}, \quad (11)$$

where B is the total bandwidth of the chirp, and $0 < k_T < 1$ and $0 < k_B < 1$ are parameters that control nonlinearity of the chirp frequency. The nonlinear portion of the chirp $\varphi(t)$ is given by

$$\varphi(t) = \mathbf{G}^{-1} \begin{bmatrix} B(1 - k_B) \\ B \\ B(1 - k_B) \\ T(1 - k_T) \end{bmatrix} \begin{bmatrix} t^2 & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (12)$$

where

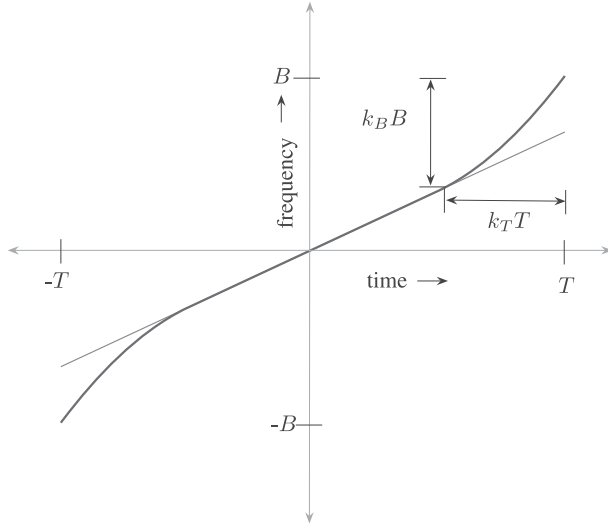


FIG. 3. Nonlinear FM characteristics.

$$\mathbf{G} = \begin{bmatrix} T^2(1 - k_T)^2 & T(1 - k_T) & 1 \\ T^2 & T & 1 \\ 2T(1 - k_T) & 1 & 0 \end{bmatrix}. \quad (13)$$

Figure 3 shows the frequency characteristics of the nonlinear FM given in (11). A major disadvantage of pulse compression waveforms are the presence of range sidelobes. Very low sidelobe levels (SLLs) are essential for weather radar applications because very strong gradients of reflectivity up to 30–40 dB km⁻¹ can occur in precipitation (Bringi and Chandrasekar 2001). The sidelobe level is often described in terms of peak sidelobe level (PSL) and integrated sidelobe level (ISL; Mudukutore et al. 1998). It is important to have low ISL because weather radars observe volume targets that extend over large areas. The range sidelobes can be reduced by the application of a compression filter that is not matched to the transmit pulse. The application of mismatched filters results in a loss of SNR because the compression filter is no longer designed to maximize signal-to-noise ratio. The NLFM waveform provides the nonlinearity parameters that can be tuned to minimize the sidelobe level along with the design of the compression filter. Such tuning parameters are unavailable with the traditional linear FM (LFM).

b. Compression filters

The range sidelobes can be reduced by using a mismatched filter. The mismatched filters are obtained by using standard window functions and least squares filters.

1) WINDOW FUNCTION FILTERS

Standard window functions have been used to shape the spectrum of the transmitted pulse to attain lower sidelobes.

f_s	Sampling rate
T	Pulse width
B	Chirp bandwidth
k_T	Non-linearity tuning factor in time
k_B	Non-linearity tuning factor in bandwidth
α_T	Tukey window parameter

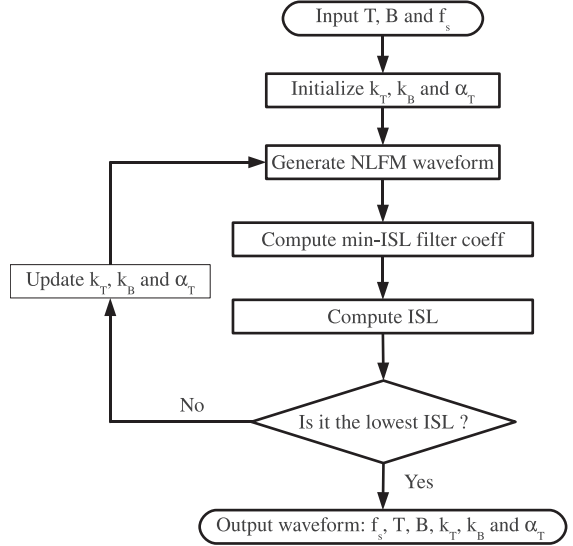


FIG. 4. Nonlinear FM pulse compression waveform specification parameters and design process.

Ideally the square root of the window functions must be applied to the transmit pulse, which results in a matched filter with low sidelobe levels. However, this would require a linear power amplifier, which is not efficient in utilizing power. Thus, window functions are applied on receive to improve sidelobe performance. The application of standard window functions, such as Taylor, Hamming, and Chebyshev weighting are readily available from literature (Cook and Bernfeld 1993). The loss in SNR is given by

$$\text{SNR}_{\text{loss}} = \frac{\left(\sum_{n=1}^N w_n \right)^2}{N \sum_{n=1}^N w_n^2}. \quad (14)$$

The loss in SNR is typically determined by the window function that is used. A very aggressive window function will have a loss in excess of 5 dB. There is a trade-off between sidelobe reduction and loss in SNR.

2) OPTIMAL ISL FILTER

Mismatched filters obtained from least squares minimization have been known to provide good performance

for some pulse compression waveforms. Ackroyd and Ghani (1973) proposed an inverse filter based on Wiener–Hopf equations, and Mudukutore et al. (1998) evaluated the applicability of inverse filter using Barker codes for weather radars. Baden and Cohen (1990) proposed the optimal ISL filter that minimized the ISL in a least squares sense, while Cilliers and Smit (2007) provided a generalization of the optimal ISL filter by extending the minimization in the L_p norm sense. The following section develops the compression filter using the L_2 norm (least squares).

Let $\mathbf{G}$ be the transmit convolution matrix obtained from the discrete complex envelope of the transmit pulse $\mathbf{g}$ and let $\mathbf{h}$ be the finite impulse response (FIR) filter coefficients of the required compression filter. The output of the compression filter with the transmit pulse as the input is given by

$$\mathbf{y} = \mathbf{G}^T \mathbf{h}, \quad (15)$$

where $\mathbf{y}$ is the output of the compression filter. Let $\mathbf{G}_m$ be the modified transmit convolution matrix obtained by deleting the columns of $\mathbf{G}$ that corresponds to the main lobe of the ambiguity function obtained from $\mathbf{y}$. The ISL is considered as the cost function that has to be minimized to obtain the FIR filter coefficients. The ISL is given by

$$\text{ISL} = f(\mathbf{h}) = \mathbf{h}^H \mathbf{G}_m^* \mathbf{G}_m^T \mathbf{h}. \quad (16)$$

The ISL is minimized by using the method of Lagrange multipliers with the constraint on the peak of $\mathbf{y}$ as $\mathbf{g}^T \mathbf{h} = \alpha$,

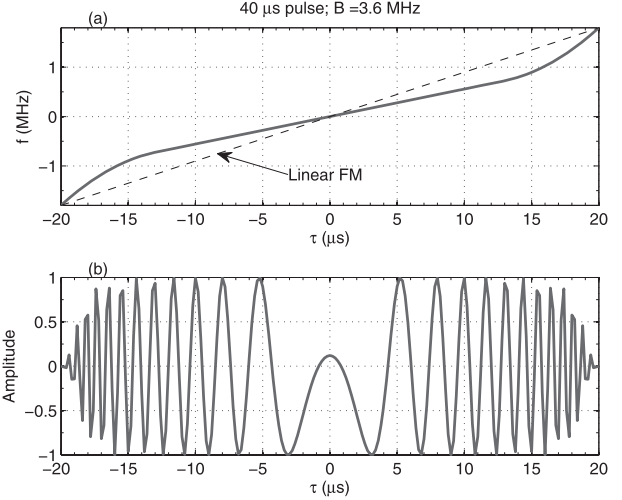


FIG. 5. Nonlinear FM pulse compression waveform (a) FM characteristics and (b) amplitude of transmit pulse. The pulse compression waveform has a nonlinear chirp of $B = 3.6$ MHz, a $T = 40$ μs pulse length, and a base-band sampling frequency $f_s = 5$ MHz.

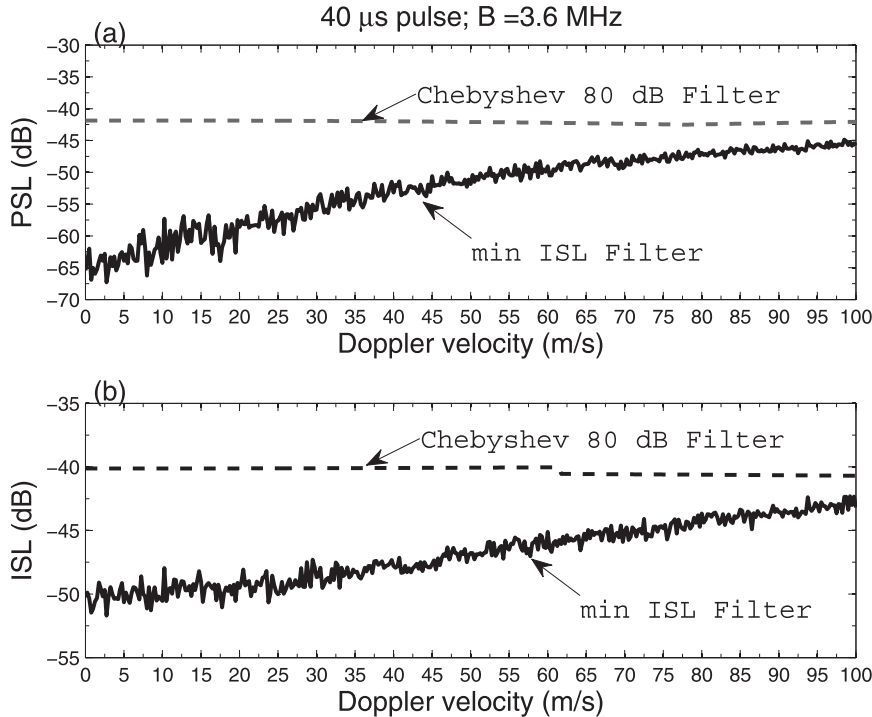


FIG. 6. Sidelobe level as a function of Doppler velocity: (a) peak sidelobe level and (b) integrated sidelobe level. The pulse compression waveform has a nonlinear chirp of $B = 3.6$ MHz, a $T = 40$ μs pulse length, and a base-band sampling frequency of $f_s = 5$ MHz.

where α is an arbitrary constant. The equation to be solved is obtained from the Lagrangian given as

$$\frac{\partial f(\mathbf{h})}{\partial \mathbf{h}^*} + \lambda \frac{\partial \Re(\mathbf{g}^T \mathbf{h} - \alpha)}{\partial \mathbf{h}^*} = \mathbf{0}. \quad (17)$$

The closed-form solution for the above equation is given as

$$\mathbf{h} = \frac{\alpha(\mathbf{G}_m^* \mathbf{G}_m^T)^{-1} \mathbf{g}^H}{\mathbf{g}(\mathbf{G}_m^* \mathbf{G}_m^T)^{-1} \mathbf{g}^H}. \quad (18)$$

The minimum ISL compression FIR filter is normalized to have unity gain at zero frequency and is given by

$$\mathbf{b} = \frac{\mathbf{h}}{|\langle \mathbf{1}, \mathbf{h} \rangle|}, \quad (19)$$

where $\mathbf{1}$ is a vector whose elements are 1 and $\langle \dots \rangle$ represents the inner product.

4. Sidelobe characterization for NLFM pulse compression

The performance of the pulse compression waveforms depends on many factors, such as the BT , k_T , k_B , and the compression filter used. For a given value of B and T there are many waveforms that can be designed by varying k_T and k_B . In general, the chirp bandwidth B is limited by the base-band sampling frequency as dictated by the sampling theorem, and the pulse lengths are selected based on the hardware limitations and sensitivity mapping, as described in section 2. The envelope $u(t)$ is selected such that there are no sharp rise time that could introduce ringing in the transmitted spectrum. A Tukey window is proposed because a single tunable parameter α_T controls the rise and fall times of the pulse envelope. The tuning parameter is part of the specification of the transmit waveform. The parameters that completely specify the waveforms are shown in Fig. 4, along with the design process used to obtain the waveform parameters. Hence, the performance of the waveform is governed by three tunable parameters: α_T , k_T , and k_B . Figure 5 shows a waveform with a chirp bandwidth $B = 3.6$ MHz and $T = 40 \mu\text{s}$ operating at a base-band sampling frequency $f_s = 5$ MHz. The waveform parameters are $\alpha_T = 0.127$, $k_T = 0.354$, and $k_B = 0.6$.

a. Doppler tolerance

The main drawback of the minimum ISL filter is limited Doppler tolerance, while the window function filter is known to be Doppler tolerant. To compare the performance of the compression filter to the Doppler shifts, the received signal is shifted in frequency based on the mean Doppler velocity of the resolution volume. The increase in sidelobe level due to Doppler shift is shown

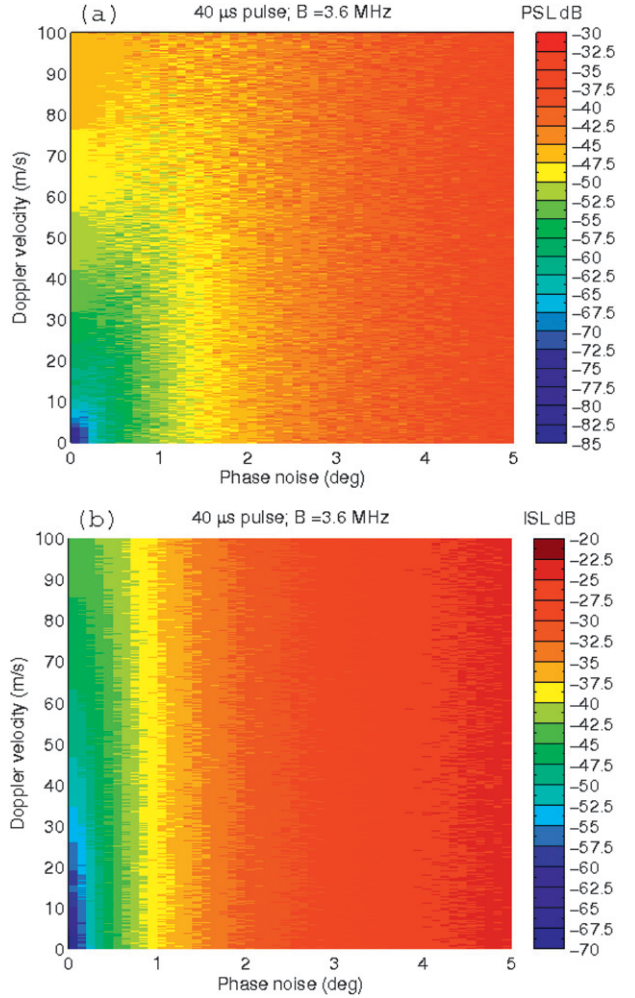


FIG. 7. Sidelobe level as a function of Doppler velocity and system phase noise: (a) peak sidelobe level and (b) integrated sidelobe level. The pulse compression waveform has a nonlinear chirp of $B = 3.6$ MHz, a $T = 40 \mu\text{s}$ pulse length, and a base-band sampling frequency of $f_s = 5$ MHz.

as a function of mean velocity in Fig. 6. The peak sidelobe level and integrated sidelobe level are almost invariant with velocity for the window function, while both the peak sidelobe level and integrated sidelobe level increase for the minimum ISL filter as shown in Figs. 6a,b. However, it is important to note that although the sidelobe level increases with velocity the performance of the minimum ISL filter is much better than the 80-dB Chebyshev window filter. The ISL of the minimum ISL filter is better than -45 dB for Doppler velocities less than 50 m s^{-1} .

b. Phase noise tolerance

The nonlinear FM is implemented as a digital FM signal and upconverted to the radio frequency (RF). The phase on the transmitted subpulses has random phase errors because of the phase noise of the system. The

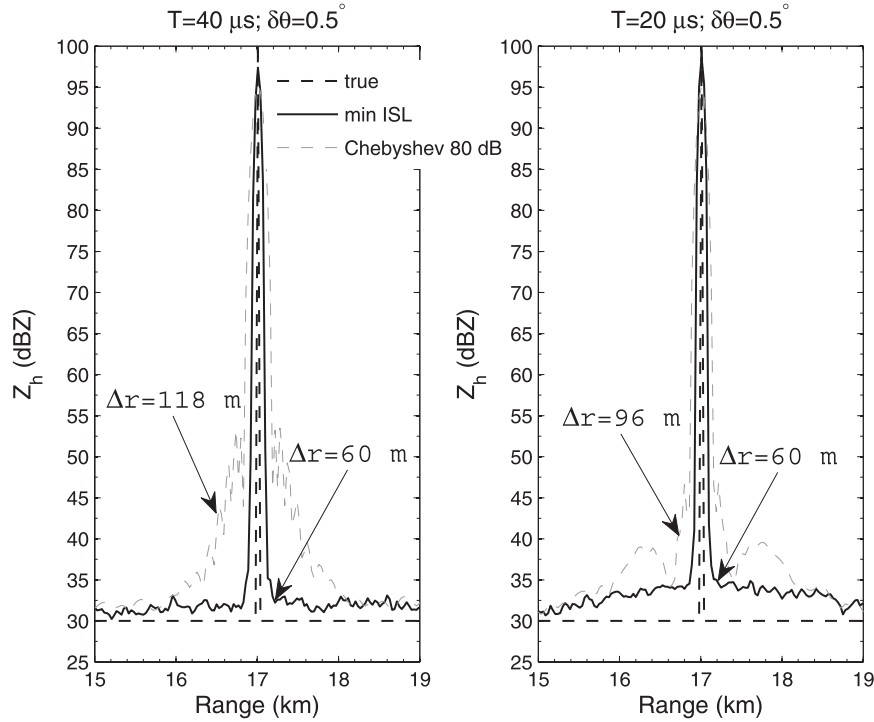


FIG. 8. Comparison of range resolution using a Chebyshev 80-dB mismatched filter and a minimum ISL filter for a pulse compression waveform using $B = 3.6$ MHz, (left) $T = 40 \mu\text{s}$ and (right) $T = 20 \mu\text{s}$ pulse length, and a base-band sampling frequency of $f_s = 5$ MHz.

individual elements in the radar system, such as Stable Local Oscillator (STALO), mixers, and transmitters, contribute to the system's phase noise. The system's phase noise has an impact on the performance of the pulse compression waveform.

The PSL and ISL as a function of phase noise are shown in Fig. 7. It can be observed that both PSL and ISL decrease with an increase in phase noise. A phase noise of 0.25° is required to achieve an ISL better than 50 dB, and phase noise of 0.5° is required to achieve an ISL better than 40 dB. Therefore, phase noise is an important factor that must be taken into consideration to ascertain the performance of a pulse compression radar.

5. Pulse compression for volume targets

The impact of the phase noise, Doppler velocity, and choice of filter was described in the previous sections. In this section we describe the resolution and receiver bandwidth of the proposed pulse compression waveform for volume targets. In addition, the effect of the sidelobe levels on the estimated Doppler spectral moments while observing volume targets is presented.

a. Range resolution

In a traditional pulsed Doppler weather radar the range resolution is determined by the transmitted pulse

width. However, the range resolution with pulse compression waveforms is determined by the chirp bandwidth B and compression filter. For a matched filter the range resolution is $c/2B$, where c is the speed of light. The range resolution with a pulse compression waveform is obtained by simulating a point target and calculating the effective pulse width after the compression (Brangi and Chandrasekar 2001). Figure 8 shows a comparison of the range resolution using a minimum ISL filter and Chebyshev 80-dB mismatched filter. The comparison of range resolution is shown for the pulse compression waveform using $B = 3.6$ MHz, $T = 40 \mu\text{s}$, and $T = 20 \mu\text{s}$ pulse length. The minimum ISL filter provides a range resolution $\Delta r = 60$ m for both $T = 40 \mu\text{s}$ and $T = 20 \mu\text{s}$, while the Chebyshev 80-dB mismatched filter provides a range resolution of $\Delta r = 118$ m and $\Delta r = 96$ m for $T = 40 \mu\text{s}$ and $T = 20 \mu\text{s}$, respectively. The minimum ISL filter provides better range resolution than the Chebyshev 80-dB mismatched filter in addition to providing better range sidelobe suppression.

b. Receiver bandwidth

The spectrum of the transmitted pulse is shown in Fig. 9a along with the filter characteristics of the minimum ISL filter and the 80-dB Chebyshev window mismatched

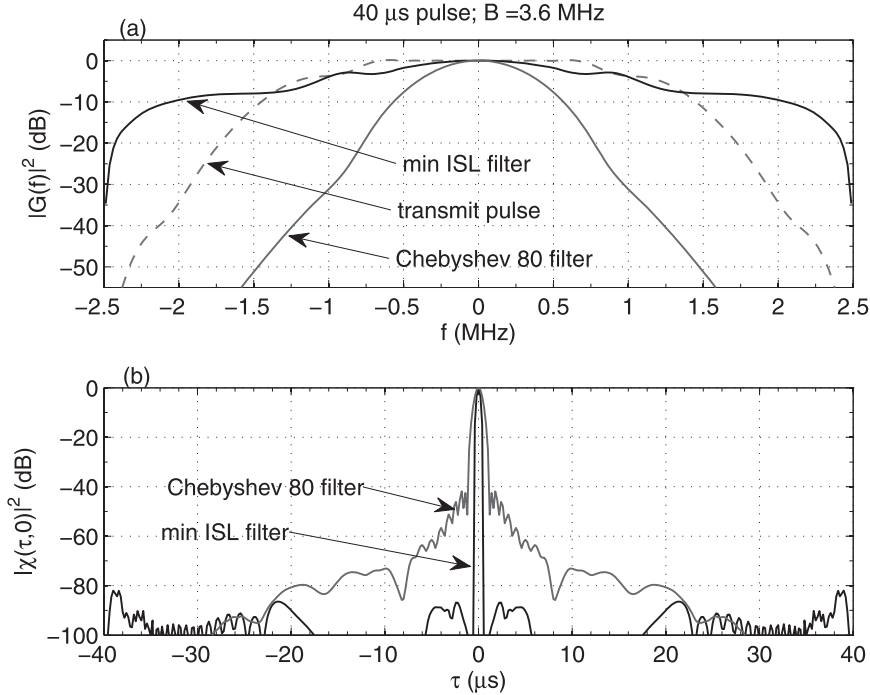


FIG. 9. (a) Spectrum of transmit pulse and compression filter characteristics and (b) comparison of ambiguity function with 80-dB Chebyshev window filter and minimum ISL filter. The pulse compression waveform has a nonlinear chirp of $B = 3.6$ MHz, $T = 40$ μ s pulse length, and a base-band sampling frequency of $f_s = 5$ MHz.

filter. It can be observed that the minimum ISL filter retains most of the frequency components, while the 80-dB Chebyshev window filter has a much narrower bandwidth. Both filters reduce the sidelobe level, as shown in Fig. 9b. However, the minimum ISL filter has a much lower sidelobe level when compared to the window function. This is an obvious result because the minimum ISL filter was designed to minimize the total energy in the sidelobes. The bandwidth of the receiver is finite and this finite bandwidth results in the loss of received power because some of the spectral components of the received signal will be filtered out. It is important to have a filter that does not have a large finite bandwidth loss (ℓ_r). The finite bandwidth loss must be taken into account while estimating a calibrated reflectivity factor (Z_h).

Table 1 lists the finite bandwidth loss for various compression filters applied to the proposed pulse compression waveform. The minimum ISL filter has the least finite bandwidth loss of $\ell_r = 2.49$ dB for $T = 20$ μ s pulse and $\ell_r = 1.83$ dB for $T = 40$ μ s. The mismatched filter using window functions, which provide relatively lesser finite bandwidth loss, do not have very good performance in terms of sidelobe suppression. The Chebyshev 80-dB mismatched filter provides good sidelobe suppression but, because of its narrower bandwidth, it also has a larger

finite bandwidth loss of $\ell_r \geq 4.5$ dB. The minimum ISL filter has lower finite bandwidth loss when compared to a window-based mismatched filter, in addition to providing better range sidelobe suppression.

c. Impact on Doppler spectral moments

A simulation is performed to evaluate the impact of SLL. The simulations of weather echoes are based on the methodology presented by Chandrasekar et al. (1986), and this methodology is used to simulate the received signal from the pulse compression waveform as described by Mudukutore et al. (1998). A trapezoidal profile of reflectivity is used to simulate the range profile. The height and gradient of reflectivity can be controlled to evaluate the impact of the reflectivity variation along the range.

TABLE 1. Finite bandwidth filter loss (dB) for frequency diversity pulse compression waveform $B = 3.6$ MHz.

Filter	$T = 20$ μ s, $\alpha_T = 0.1944$	$T = 40$ μ s, $\alpha_T = 0.1268$
	Filter loss (ℓ_r)	Filter loss (ℓ_r)
Min ISL	2.49	1.83
Hamming	3.47	3.71
Hann	3.70	3.92
Chebyshev 80 dB	4.65	4.88

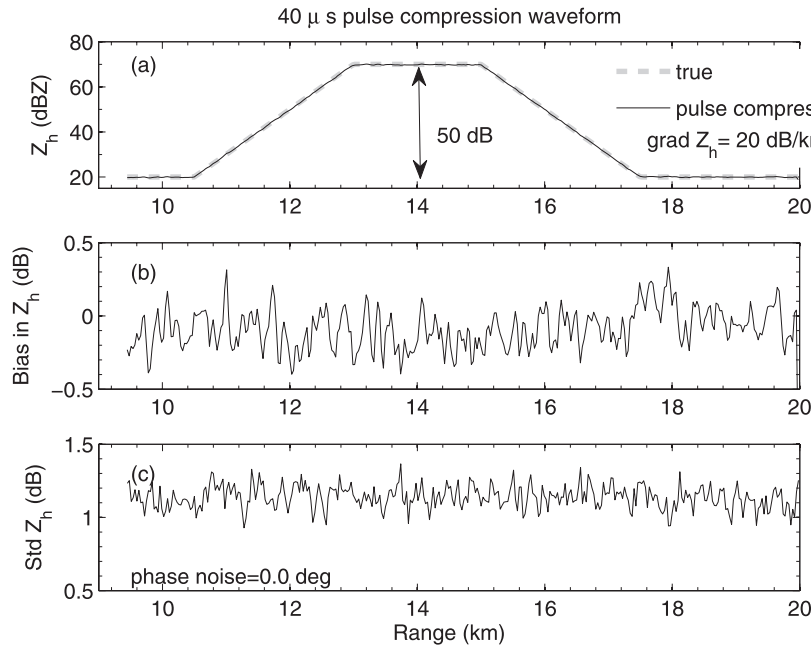


FIG. 10. Observation of a trapezoidal reflectivity profile with 20 dB km^{-1} gradient. (a) Observed mean reflectivity, (b) bias in measured reflectivity, and (c) standard deviation of reflectivity. The pulse compression waveform has a nonlinear chirp of $B = 3.6 \text{ MHz}$, $T = 40 \text{ }\mu\text{s}$ pulse length, and a base-band sampling frequency of $f_s = 5 \text{ MHz}$. The phase noise of the system is 0° and a reflectivity ramp height of 50 dB .

Figure 10 shows the simulation results for a $T = 40 \text{ }\mu\text{s}$ pulse compression waveform with $N = 64$ pulses. The simulations were carried out for a spectral width of $\sigma_v = 2 \text{ m s}^{-1}$ using a PRF of 2 kHz . Simulations

were repeated 100 times for the same reflectivity profile. The dashed gray line in Fig. 10a is the true profile, with a reflectivity gradient of 20 dB km^{-1} . The floor of the profiles is well above the noise floor. Figure 10a

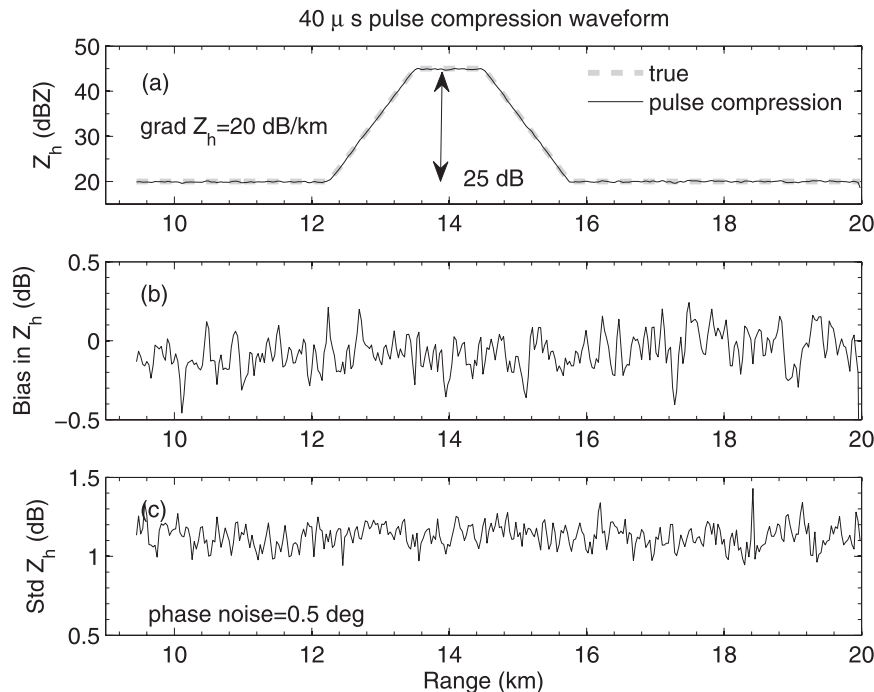


FIG. 11. As in Fig. 10, but the phase noise of the system is 0.5° , and the reflectivity ramp height is 25 dB .

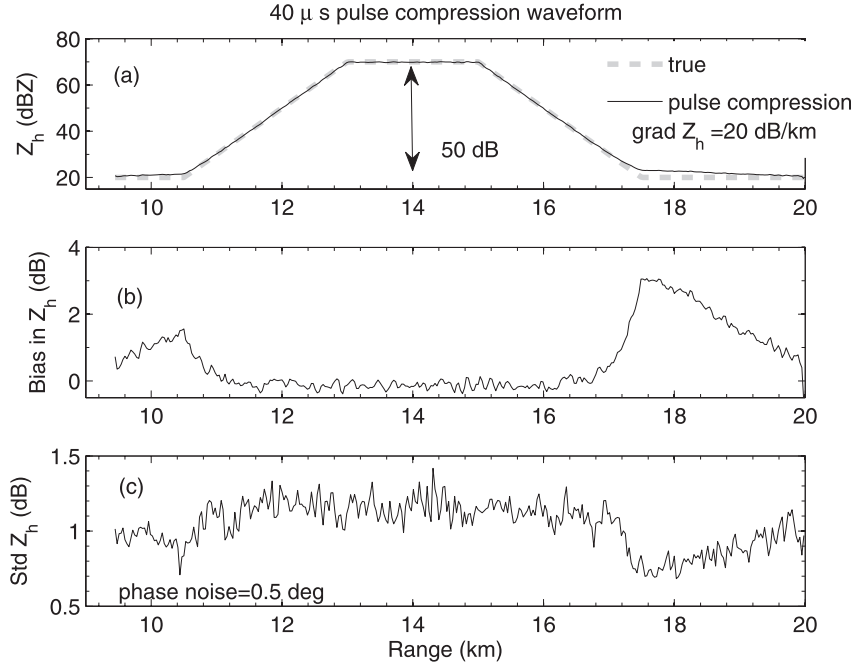


FIG. 12. As in Fig. 10, but the phase noise of the system is 0.5° , and the reflectivity ramp height is 50 dB.

shows the estimated reflectivity for an ideal system with zero phase noise. It can be observed that ISL does not bias the reflectivity estimates more than 0.5 dB (Fig. 10b), and the standard deviations are less than 1.5 dB.

However, the presence of phase errors will lead to degraded performance in sidelobe levels. A simulation with the same parameters are described above is performed, but with a phase noise of 0.5° . Figures 11 and

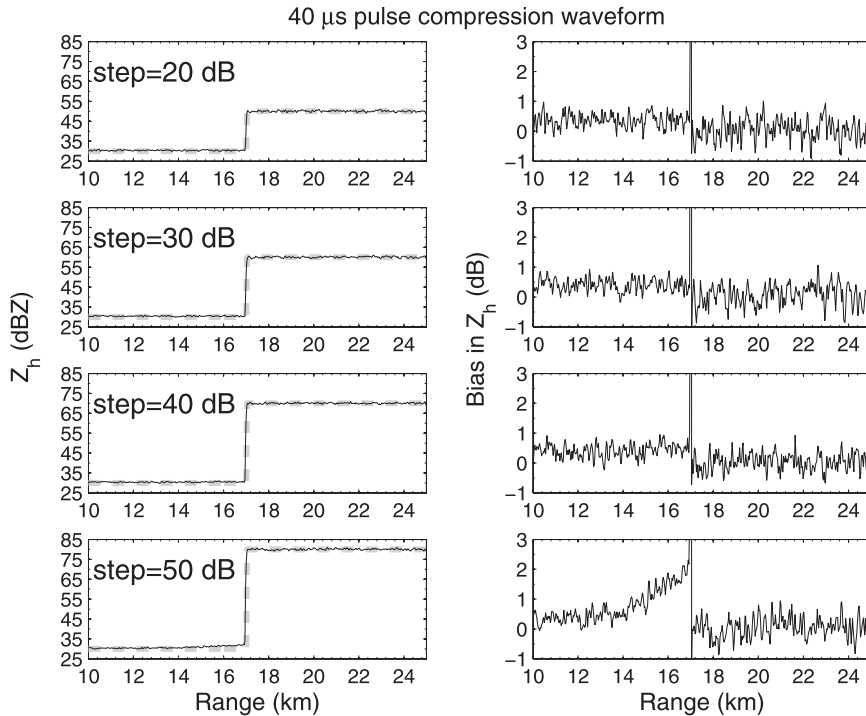


FIG. 13. The impact of sidelobe level on (left) Z_h using a step function profile with varying step sizes of reflectivity with phase noise $\delta\theta = 0.25^\circ$ and (right) its bias.

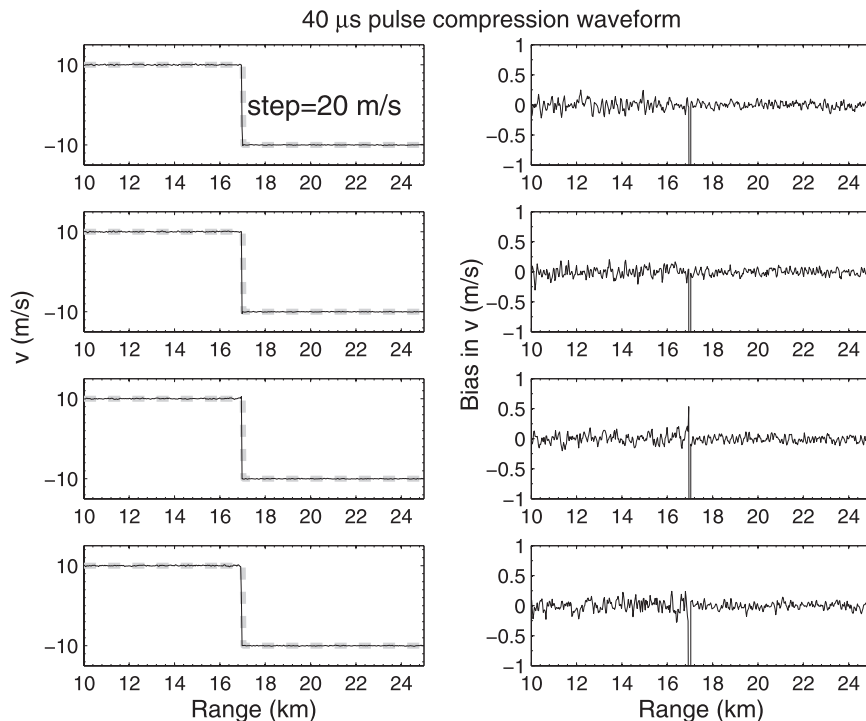


FIG. 14. The impact of sidelobe level on (left) v using a step function profile with varying step sizes of reflectivity with phase noise $\delta\theta = 0.25^\circ$ and (right) its bias.

12 show the estimated reflectivity and error in the estimated reflectivity for two different reflectivity profiles. The gradient of reflectivity is 20 dB km^{-1} for both the cases.

It can be observed from Figs. 11 and 12 that a change of 25 dB in range does not bias the reflectivity estimates, but a change of 50 dB (Fig. 12a) introduces a bias greater than about 3 dB because of the sidelobes. Therefore, phase noise plays an important role in governing the performance of pulse compression waveform.

Very high gradients in reflectivity (Z_h) within the long pulse can occur at the edges of strong convective cells and regions where there is clutter contamination. A step function with varying step sizes is simulated to analyze the limitation of the pulse compression waveform with a phase noise of $\delta\theta = 0.25^\circ$ operating at X band and PRF = 2 kHz. The floor of the step function is set at 30 dBZ so that a low signal-to-noise ratio is avoided. The step size is increased from 20 to 50 dB in increments of 10 dB. The mean Doppler velocity is set to two levels: one at 10 m s^{-1} in the 30-dBZ region and the other at -10 m s^{-1} in the stepped reflectivity region. The impact of the step size on the observed reflectivity and mean Doppler velocity is shown in Figs. 13 and 14, respectively, for $T = 40 \mu\text{s}$. Figure 13 shows the input profiles (left panel) and the bias in the estimated reflectivity (right panel). The bias in reflectivity is within acceptable levels

of $\pm 0.5 \text{ dB}$ for a step size up to 40 dB, but the impact of the sidelobe is clearly seen as a biased reflectivity estimate for a step size of 50 dB. Similar results were obtained for $T = 20 \mu\text{s}$ but are not shown in this paper. The ISL of at least 40 dB is required to handle large gradients in reflectivity and provide estimated Doppler moments with acceptable bias.

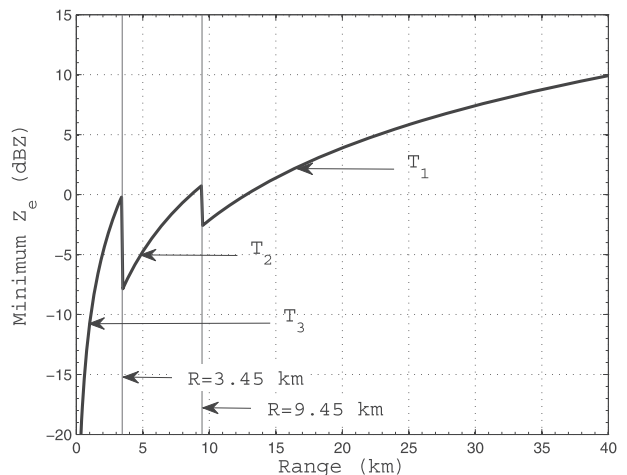


FIG. 15. Minimum detectable reflectivity of a sensitivity-mapped frequency diversity waveform with three components. The peak transmit power $P_t = 100 \text{ W}$ operates at X band.

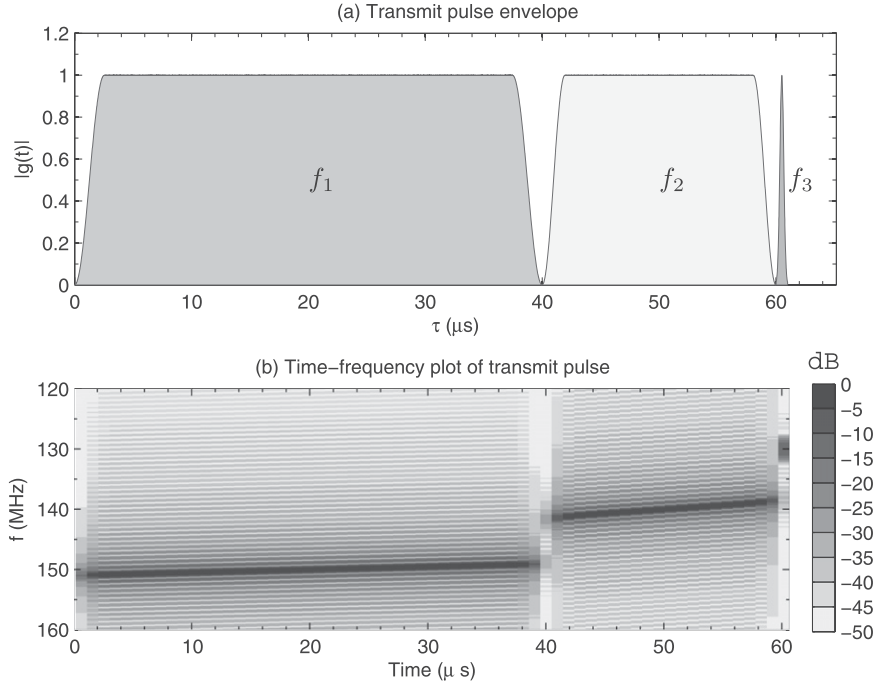


FIG. 16. Time-frequency plot of the frequency diversity pulse compression waveform: (a) transmit pulse envelope and (b) time-frequency plot.

6. Frequency-diverse waveform for WiBEX radar

Colorado State University is developing a Wideband Experimental X-band (WiBEX) radar that uses a solid-state transmitter. The main characteristics of WiBEX are shown (Table 3). Based on the requirement of the 10-dBZ sensitivity at 40 km for WiBEX, and a sensitivity of no greater than 0 dBZ at 5 km, a frequency-diverse waveform consisting of three subpulses is obtained using the sensitivity-mapped generalized waveform as shown in Fig. 15. It can be observed in Fig. 15 that a two-subpulse waveform could not have met the sensitivity requirement. A similar waveform is described by Carswell et al. (2008). The pulse widths of the three components are $T_1 = 40 \mu\text{s}$, $T_2 = 20 \mu\text{s}$, and $T_3 = 1 \mu\text{s}$. The pulse compression waveform for each component using pulse compression is obtained by minimizing the sidelobe for a $B = 3.6 \text{ MHz}$ chirp operating at a base-band sampling frequency of $f_s = 5 \text{ MHz}$. The minimum ISL filter is chosen as the compression filter. The use of frequency diversity enables the mitigation of blind range. The envelope of the transmit pulse and its time-frequency plot are shown in Fig. 16. The parameters of the frequency-diverse waveform are shown in Table 2. The subpulse at frequency f_1 has a blind range of 9.15 km and has the highest sensitivity (as shown in Fig. 1), while the subpulse at frequency f_2 has a blind range of 3.15 km and has intermediate sensitivity. The short pulse at f_3 does not

have a blind range and has the least sensitivity. The measurements from the three frequencies are combined to provide observations without any blind range and adequate sensitivity.

To ascertain the feasibility of the frequency-diverse pulse compression waveform in a more realistic meteorological phenomenon a simulation based on observations from CSU-CHILL radar is performed. A precipitation event with a well-defined bright band was observed by the CSU-CHILL radar at 0215:01 UTC 7 June 2003. This brightband dataset was used to simulate the received signal for a frequency-diverse pulse compression waveform transmitted using a 100-W solid-state power amplifier. A phase noise of 0.5° was added to the transmit waveform. The simulations were performed with $N = 64$ pulses at a PRF of 2 kHz. The calibration was done based on the radar constant for each component of the transmit waveform. The true brightband observations are shown

TABLE 2. Frequency diversity pulse compression waveform.

Parameter	$i = 1$	$i = 2$	$i = 3$
Frequency	f_1	f_2	f_3
Pulse width (μs)	40	20	1
Chirp bandwidth (MHz)	3.6	3.6	—
Tukey window tuning	0.1268	0.1944	—
k_T	0.3543	0.3274	—
k_B	0.6	0.4920	—

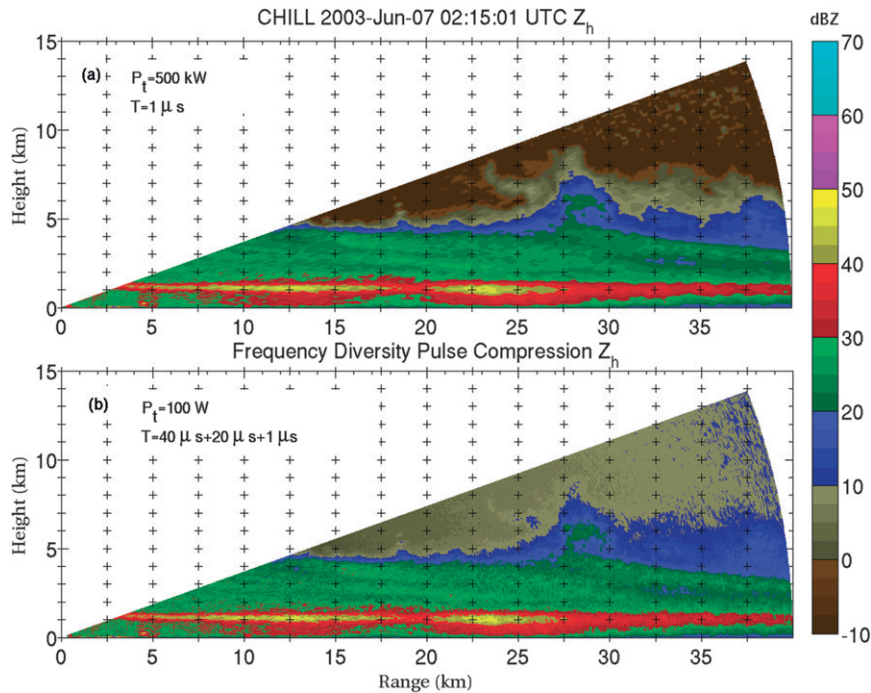


FIG. 17. Simulation of frequency diversity pulse compression waveform for an X-band radar. (a) Observed reflectivity field from the CSU-CHILL radar at 0215:01 UTC 7 Jun 2003 and (b) the retrieved reflectivity from a frequency diversity pulse compression waveform.

in Fig. 17a, which was obtained from a 500-kW peak-power radar. The reflectivity obtained by combining the observations from the three components of the transmit pulse is shown in Fig. 17b. A comparison of the true reflectivity and that obtained from pulse compression agree well. The reflectivity observations with the solid-state radar and true reflectivity deviate from each other at lower reflectivity regions seen well above the melting layer at farther ranges, as shown in Fig. 17. This difference is primarily because CSU-CHILL uses a very high powered Klystron transmitter and has much better sensitivity.

A performance of the frequency-diverse pulse compression radar is evaluated based on simulations at X band. The characteristics of the solid-state radar used in the simulations are shown in Table 3. Observations from the X-band polarimetric radars deployed by the Center for Collaborative Adaptive Sensing of the Atmosphere (CASA) are used as input data to simulate a more realistic distribution of precipitation, and simulations were performed with $N = 64$ pulses at a PRF of 2 kHz. The simulations are carried out at three different frequencies whose center frequencies are separated by 10 MHz.

The observations made by the CASA radar at Chickasha, Oklahoma, at 0414:19 UTC 10 March 2009, and the results obtained with the frequency-diverse

pulse compression waveform are shown in Fig. 18. The observations from CASA are clutter filtered at an elevation angle of 2° (Bharadwaj and Chandrasekar 2010). The precipitation event shown in Fig. 18 has weaker echoes close to the radar from an azimuth of 240° to 330° and very strong reflectivities in the ranges covered by the longest pulse $T_1 = 40 \mu\text{s}$ (an azimuth of 330° – 20°). The results obtained for the frequency-diverse pulse compression waveform are combined from the three

TABLE 3. Solid-state radar characteristics used in simulations.

Transmitter	
Type	Solid state
Center frequency	9400 ± 100 MHz
Peak power output	100 W (per polarization channel)
Pulse width	Maximum $70 \mu\text{s}$
Polarization	Dual linear, horizontal and vertical
Max duty cycle	15%
Antenna and pedestal	
Type (diameter)	Parabolic reflector (2.4 m)
3-dB beamwidth	1°
Gain	43.0 dB
Receiver	
Type	Dual-channel digital
Bandwidth	5 MHz (before compression filter)
Noise figure	4.0 dB
Sampling rate	5 MHz (baseband)

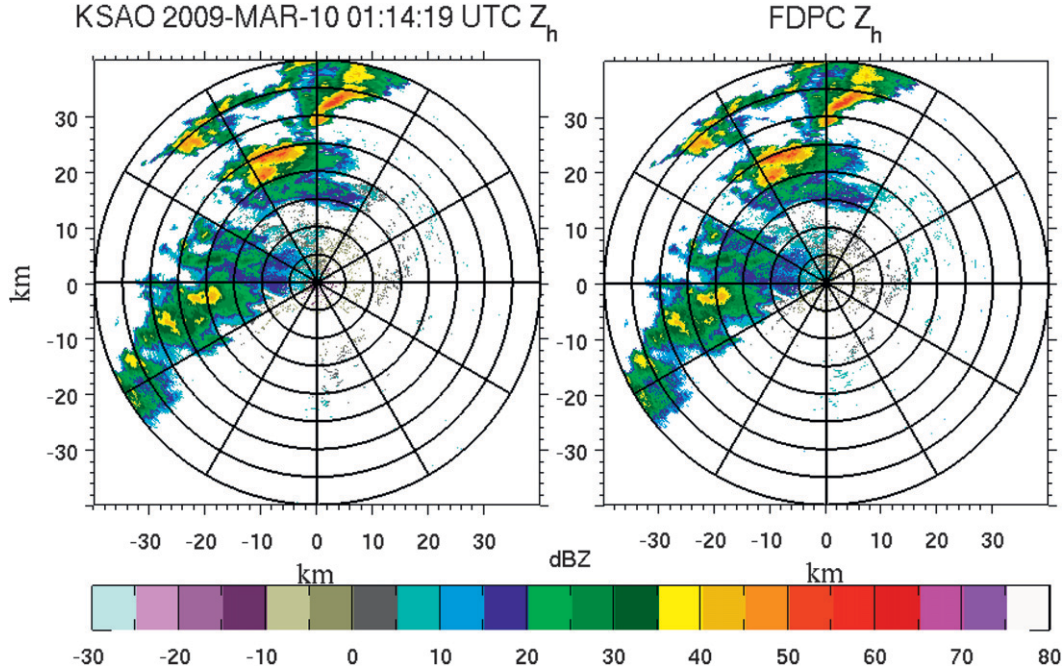


FIG. 18. (right) Plan position indicator plots of frequency diversity pulse compression waveform compared with (left) the input data. The simulation is based on observation made with CASA IP1 radar at Chickasha at 0114:19 UTC 10 Mar 2009.

frequencies such that there is no blind range in the data and there is spatial continuity along range. The reflectivity observed with the frequency-diverse pulse compression is shown in Fig. 18, and it can be seen that they match very well with the input reflectivity. The reflectivity close to the radar is obtained from the short pulse while the weaker echoes farther away from the radar are obtained from the long pulses. Similar simulations were carried out for a second dataset based on observations made by the CASA radar at Cyril, Oklahoma, at 2314:17 UTC 16 April 16 2009 under identical setup. The simulation based on Cyril radars also provided results that are comparable to the Chickasha results. A scatterplot of reflectivity observed using frequency-diverse pulse compression versus input reflectivity for the two simulation cases is shown in Fig. 19. The scatterplot is segmented according to varying radar range regions (e.g., 0–4 and 4–8 km), and the reflectivities obtained from frequency-diverse pulse compression are successively offset by 20 dB for each range region. The offset is added only for the scatterplot to observe and compare the results of the simulation. The minimum observable reflectivity is also plotted on the scatterplot. The scatterplots show that the input reflectivities and simulated observations agree with each other above the minimum observable reflectivity. The observations from frequency-diverse pulse compression below the minimum observable reflectivity is dominated mostly by the receiver noise and observations

that no longer maintain a one-to-one relation with the input reflectivities, as seen in the scatterplot.

7. Summary

Waveforms for radar using solid-state transmitters are becoming an important part of the strategy for electronically scanned radars. The main drawback of the solid-state transmitter is the unavailability of high peak power, which reduces the sensitivity. However, solid-state transmitters are capable of transmitting long pulses, which are used to gain sensitivity. The transmission of long pulses does provide good sensitivity, but at the expense of very poor range resolution. Pulse compression waveforms provide the means to achieve good sensitivity by transmitting long pulse and still making observations with good range resolution. However, the transmission of long pulses blinds the radar for the duration of the pulse length, resulting in blind ranges. A novel waveform using frequency diversity is presented to mitigate the problems of low sensitivity and blind ranges associated with pulse compression waveforms. A sensitivity-mapped generalized waveform was presented that utilized wideband transmit signals along with frequency diversity.

A NLFM pulse compression waveform was proposed for the long pulses to achieve good range resolution and sidelobe levels. The nonlinear frequency modulation

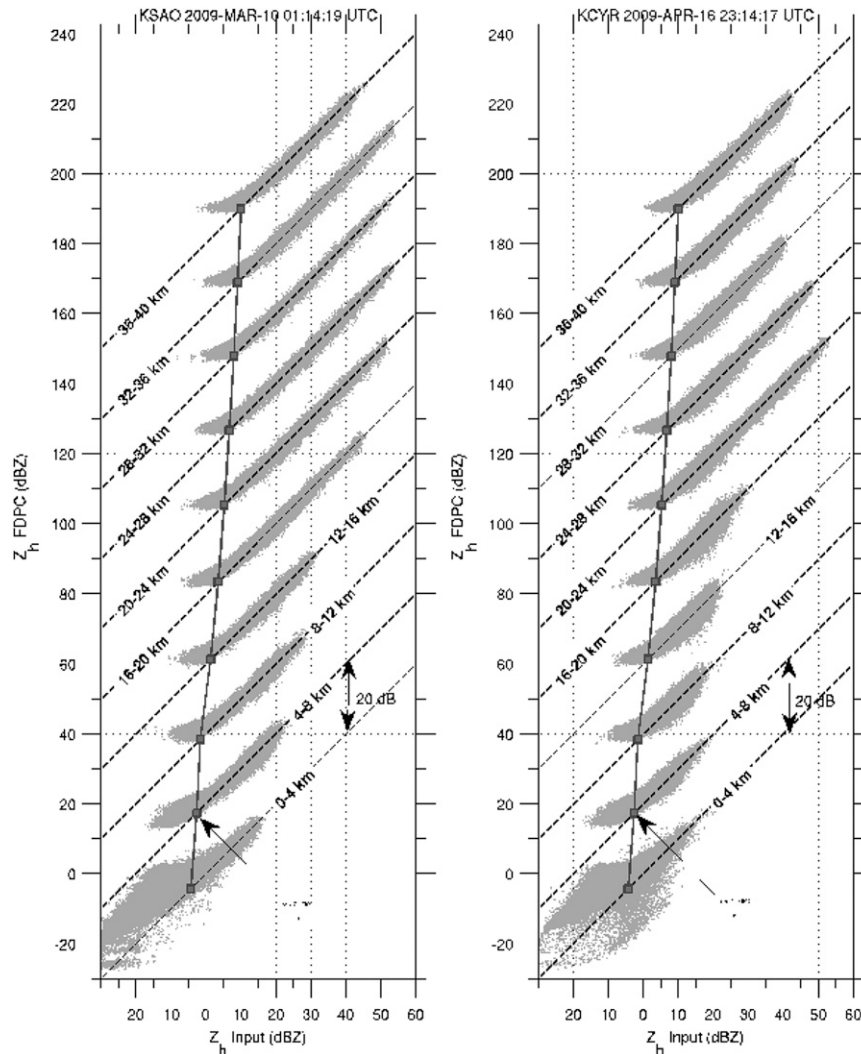


FIG. 19. Scatterplot of reflectivity from frequency diversity pulse compression vs the input reflectivity. The data used for simulations were collected by (left) the Chickasha radar at 0114:19 UTC 10 Mar 2009 and (right) by the Cyril radar at 2314:17 UTC 16 Apr 2009.

pulse compression waveform was parameterized in terms of a linear and a nonlinear frequency chirp segment. A quadratic time–frequency relation governs the nonlinear frequency modulation. The pulse compression waveform is controlled by three parameters—the time and bandwidth control parameters and the Tukey envelope window parameter. A minimum ISL filter was designed to achieve very good sidelobe performance. The minimum ISL filter is based on the transmit pulse and is designed such that the ambiguity function has minimal integrated sidelobes. The sidelobe performance of pulse compression waveforms were presented for the minimum ISL filter. The Doppler tolerance and the impact of phase errors were quantified for integrated and peak sidelobe levels. It is important to have phase

errors less than 0.5° to minimize the range sidelobe problem. The performance of the minimum ISL filter degrades in the presence of Doppler velocity, but the sidelobe levels are still within acceptable limits for Doppler velocities encountered in meteorological phenomena.

An analysis of the pulse compression waveform for volume targets was presented. The range resolution, receiver, and impact on the estimated Doppler moments were presented for the pulse compression waveform. The minimum ISL filter provided better range resolution when compared to the window-based mismatched filters. The receiver bandwidth plays an important role in the amount of added noise and finite bandwidth loss in the receiver. The minimum ISL filter suffers less finite bandwidth loss when compared to the window-based

mismatched filters. The impact of strong reflectivity gradients and range sidelobes were presented for volume targets. The impact of range sidelobes using step functions with varying step sizes was presented. The retrieval of the Doppler moments was not significantly affected by a reflectivity step size up to 40 dB. The performance of the nonlinear pulse compression waveform and the minimum ISL filter were acceptable for the reflectivity step size up to 40 dB.

A frequency-diverse pulse compression waveform was simulated based on actual observations from the CSU–CHILL radar at S band and the CASA Integrated Project 1 (IP1) radars at X band. The CSU–CHILL radar observations are made with a very high power transmitter and, therefore, have very high sensitivity. A comparison based on CSU–CHILL radar observations with the simulated X-band observation using frequency-diverse pulse compression waveform was presented. The frequency-diverse pulse compression waveform does provide adequate sensitivity and is suggested for operations. The performance of the frequency-diverse pulse compression waveform based on observations from CASA's IP1 radars was presented. The errors in Z_h and mean velocity were described for varying reflectivity ranges. The biases and standard deviations are within acceptable limits. The frequency-diverse pulse compression waveform presented in this paper provided acceptable performance in providing adequate sensitivity, minimizing range sidelobes, and mitigating the blind range. Based on analysis performed on realistic simulations using CSU–CHILL radar data and CASA IP1 data, the frequency-diverse pulse compression waveform is suggested for polarimetric pulsed Doppler weather radars using solid-state transmitters.

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